

VLADIKAVKAZ, Russian Federation

WMO: 372280

Lat: 43.033N

Lon: 44.683E

Elev: 703

StdP: 93.16

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-10.9	-16.6	1.0	-11.9	-14.2	1.2	-8.2	7.1	4.3	5.4	0.1	1.3	170	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	31.1	20.3	29.3	19.9	27.7	19.3	22.1	28.4	21.2	27.1	20.3	25.8	2.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	16.1	25.6	19.1	15.1	24.5	18.3	14.4	23.4	68.5	28.4	65.0	27.1	61.8	25.9	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.5	3.7	-16.6	33.8	3.4	2.5	-19.1	35.6	-21.1	37.0	-23.0	38.4	-25.5	40.2		
(5)			-16.9	23.7	3.2	1.6	-19.2	24.9	-21.1	25.8	-22.9	26.7	-25.3	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-1.5	-0.2	4.3	9.8	15.1	18.8	21.5	21.3	16.2	10.6	4.4	0.1		
	DBStd	9.21	4.57	5.23	4.66	4.70	3.98	3.25	2.83	3.18	3.69	4.59	4.71	4.95		
	HDD10.0	1431	356	286	187	61	6	0	0	0	3	47	177	309		
	HDD18.3	3297	614	519	437	259	113	32	5	10	81	243	418	566		
	CDD10.0	1464	0	1	9	53	164	265	356	351	190	65	8	1		
	CDD18.3	287	0	0	1	2	13	47	102	102	18	3	0	0		
	CDH23.3	2114	0	0	5	24	105	332	722	760	138	28	0	0		
	CDH26.7	591	0	0	1	6	18	76	212	240	34	5	0	0		
(14) Wind	WSAvg	1.5	1.4	1.6	1.9	1.6	1.7	1.7	1.6	1.5	1.4	1.2	1.3	1.4		
(15) Precipitation	PrecAvg	895	27	29	48	79	136	157	121	93	72	52	43	34		
	PrecMax	1438	76	84	120	149	262	245	348	378	146	136	122	117		
	PrecMin	614	2	6	6	2	39	89	12	23	12	1	11	10		
	PrecStd	177	17	16	27	36	59	44	65	61	32	26	26	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	16.0	22.9	26.3	28.9	31.5	33.1	33.4	30.3	26.7	20.0	16.3		
		MCWB	5.6	7.8	10.7	13.0	18.7	20.0	21.4	20.8	19.2	14.0	9.3	7.3		
	2%	DB	9.4	12.3	17.9	22.6	26.1	28.8	30.9	31.2	27.0	22.9	16.7	11.5		
		MCWB	3.9	5.9	8.5	12.8	16.6	19.2	20.8	20.2	18.0	14.3	8.5	5.3		
	5%	DB	6.6	9.2	14.7	20.2	24.1	26.8	29.1	29.2	24.5	19.9	13.9	8.7		
		MCWB	2.5	4.3	7.5	12.0	16.1	18.8	20.2	20.2	17.5	13.4	7.9	4.1		
	10%	DB	4.2	6.5	11.7	17.7	22.1	25.1	27.2	27.4	22.5	17.3	11.1	6.2		
		MCWB	1.4	3.0	6.5	11.1	15.5	18.1	19.5	19.7	16.8	12.4	6.8	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.4	11.5	15.1	19.9	22.6	23.3	23.0	21.0	16.5	11.2	8.1		
		MCDB	11.9	14.9	20.9	22.7	26.3	28.1	30.3	29.5	27.1	21.9	16.4	13.8		
	2%	WB	4.3	6.3	9.5	13.8	18.3	20.7	22.2	22.0	19.1	15.0	9.7	6.1		
		MCDB	8.2	11.4	16.2	20.7	24.2	26.4	28.6	28.4	24.7	20.4	14.6	10.2		
	5%	WB	3.0	4.7	8.1	12.5	16.9	19.6	21.2	21.1	18.1	14.0	8.4	4.7		
		MCDB	5.9	8.8	13.3	18.5	22.3	25.2	26.9	27.2	23.0	18.7	12.8	7.6		
	10%	WB	1.7	3.2	6.8	11.5	15.8	18.7	20.3	20.3	17.2	13.0	7.2	3.4		
		MCDB	3.9	6.2	11.0	16.8	21.2	23.8	25.6	25.8	21.6	16.7	10.5	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	8.1	8.6	8.9	8.8	8.8	9.1	8.8	8.3	7.8	7.8		
		MCDBR	11.6	13.0	14.2	12.9	12.3	11.6	11.6	12.2	12.9	12.9	12.7	11.6		
	5% WB	MCWBR	7.3	7.7	7.5	6.0	6.3	6.1	5.1	5.2	6.0	6.4	6.9	6.9		
		MCWBR	10.5	12.0	12.5	11.7	11.2	10.5	10.4	11.0	10.8	11.1	11.2	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.379	0.450	0.516	0.480	0.482	0.530	0.523	0.449	0.406	0.354	0.330		
	taud		2.217	2.079	1.938	1.845	2.014	2.065	1.955	1.965	2.135	2.192	2.271	2.248		
	Ebn at Noon		773	791	775	756	801	800	754	743	773	755	746	741		
	Edh at Noon		93	127	166	196	171	163	180	173	135	112	88	83		
(44) All-Sky Solar Radiation	RadAvg		1.59	2.45	3.22	4.28	5.20	5.88	5.85	5.37	4.12	2.67	1.77	1.35		
	RadStd		0.21	0.38	0.38	0.57	0.54	0.40	0.47	0.47	0.43	0.35	0.24	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page